AI PRODUCT MANAGEMENT 2 · DETAILED COURSE

Agentic AI and Autonomous AI Systems

How AI agents plan, use tools and act, and how to run them safely and reliably in production.

Chapter 15 of 26 · 33 topics · Edition 2026 · skillnavora.in

CHAPTER 15

Agentic AI and Autonomous AI Systems

How AI agents plan, use tools and act, and how to run them safely and reliably in production.

Introduction to Agentic AI

Systems where a model pursues a goal over several steps by planning, calling tools and reacting to results.

An agent pursues a goal over several steps: it plans, calls tools, observes the results and decides the next step. Products differ by how much autonomy they give the agent.

AI Agents vs Chatbots

A chatbot replies to messages; an agent takes actions toward a goal.

A chatbot replies to messages. An agent takes actions toward a goal, such as creating a ticket and updating a customer record. Agents need permissions and oversight that chatbots do not.

AI Agents vs Traditional Automation

Automation follows fixed rules; agents handle variable inputs and decide the next step, with less predictability.

Automation follows fixed rules and fails on unexpected inputs. Agents handle variable inputs and choose the next step, but they are less predictable, so they need evaluation and limits.

Agent Architecture

A model, instructions, tools, memory, a control loop and guardrails.

An agent combines a model, instructions, tools, memory, a control loop and guardrails. The loop repeats: decide, act, observe, until the goal is met or a limit is reached.

Single-Agent Systems

One agent with a set of tools handles the task; simpler to build and debug.

One agent with a defined set of tools handles the task. Single agents are simpler to build, test and debug, so start here unless the task clearly separates into roles.

Multi-Agent Systems

Several specialised agents coordinate, useful for separable roles but costlier and harder to control.

Several specialised agents coordinate, for example a researcher, a writer and a checker. They suit separable roles, but add cost, latency and harder failure analysis.

Agent Planning

Deciding the sequence of steps to reach a goal, revising as results arrive.

Planning decides the sequence of steps toward a goal and revises it as results arrive. Plans should be inspectable so users and developers can see what the agent intended.

Reasoning and Task Decomposition

Breaking a goal into smaller sub-tasks the agent can complete and verify.

Breaking a goal into smaller sub-tasks the agent can complete and verify. Each sub-task should have a clear success check.

Tool Calling

The agent selects and invokes tools such as search, databases, code and APIs.

The agent selects and invokes tools such as search, databases, code execution and APIs. Each tool needs a clear description, input schema and error messages the agent can understand.

Function Calling

Structured model output naming a function and arguments, executed by your code.

The model returns a structured request naming a function and its arguments. Your code runs the function and returns the result to the model.

Agent Memory

Information kept so the agent can use past steps, facts and preferences.

Memory keeps facts, steps and preferences the agent needs later. Decide what is stored, for how long and who can see it.

Short-Term and Long-Term Memory

Short-term lives in the current context; long-term is stored externally and retrieved when needed.

Short-term memory lives in the current context. Long-term memory is stored externally and retrieved when needed, such as a user's saved preferences.

State Management

Tracking progress, inputs and results across steps so runs can resume and be audited.

State tracks progress, inputs and results across steps. It lets runs resume after failure and gives auditors a record of what happened.

Agent Orchestration

Coordinating agents, tools and steps, including routing, ordering and error paths.

Orchestration coordinates agents, tools and steps, including routing, ordering and error paths. It is the control layer that makes multi-step work reliable.

Workflow Orchestration

Defining fixed or conditional flows around model calls for predictable behaviour.

A workflow defines fixed or conditional steps around model calls. Workflows give predictable behaviour and are often the right choice before full autonomy.

Agent Communication

How agents exchange messages and shared state, including protocols such as MCP for tools and A2A for agents.

Agents exchange messages and shared state. Protocols such as MCP connect agents to tools and data, and agent-to-agent protocols connect agents to each other.

Human-in-the-Loop

Humans approve, edit or take over at defined checkpoints.

Humans approve, edit or take over at defined checkpoints. Place checkpoints before irreversible or costly actions, such as sending a payment or deleting data.

Autonomous Execution

Agents act without per-step approval, appropriate only for low-risk, reversible tasks.

The agent acts without per-step approval. This suits low-risk, reversible tasks with good evaluation, such as tagging support tickets.

Agent Permissions

Least-privilege access to tools and data, scoped per task and revocable.

Agents get least-privilege access to tools and data, scoped to each task and revocable. An agent with broad access is a security incident waiting to happen.

Agent Error Handling

Detecting tool failures, bad outputs and loops, and responding safely.

Handle tool failures, bad outputs and loops. Set timeouts, validate outputs and return clear errors to the agent so it can recover or stop safely.

Retry and Recovery Mechanisms

Retries with backoff, fallbacks, checkpoints and idempotent actions so failures do not cause harm.

Use retries with backoff, fallbacks, checkpoints and idempotent actions. Idempotent actions are safe to repeat, so a retry does not create a duplicate payment.

Agent Evaluation

Measuring task success, steps, cost and safety on realistic scenarios.

Measure task success, number of steps, cost and safety on realistic scenarios. Include tasks that should be refused, not only tasks that should succeed.

Agent Observability

Traces of every model call, tool call and decision for debugging and audit.

Traces record every model call, tool call and decision. Traces let engineers debug failures and let auditors reconstruct what an agent did.

Agent Security

Defending against prompt injection, data exfiltration, excessive permissions and malicious tools.

Defend against prompt injection through tool outputs, data exfiltration, excessive permissions and malicious tools. Treat every external input as untrusted.

Agent Governance

Policies for what agents may do, approval rules, logging, ownership and incident response.

Governance sets what agents may do, approval rules, logging, ownership and incident response. Assign a named owner to each production agent.

LangChain

A framework for building LLM applications with chains, tools and retrievers.

LangChain is a framework for LLM applications, with chains, tools and retrievers. It speeds up prototyping, but check its abstractions against your needs before adopting it widely.

LangGraph

A library for stateful, graph-based agent workflows with loops and human checkpoints.

LangGraph builds stateful, graph-based agent workflows with loops and human checkpoints. It suits agents that must follow a defined but branching process.

CrewAI

A framework for teams of role-based agents collaborating on tasks.

CrewAI is a framework for teams of role-based agents collaborating on tasks. It makes multi-agent designs quick to prototype, so test whether the roles add value.

LlamaIndex

A framework centred on data ingestion, indexing and retrieval for LLM apps.

LlamaIndex focuses on data ingestion, indexing and retrieval for LLM applications. It is a strong choice when the main challenge is connecting models to documents.

OpenAI Agents SDK

OpenAI's SDK for building agents with tools, handoffs and guardrails.

OpenAI's SDK builds agents with tools, handoffs between agents and guardrails. Handoffs let one agent pass the conversation to a specialist.

Microsoft Agent Framework

Microsoft's framework for building and orchestrating agents, evolved from Semantic Kernel and AutoGen.

Microsoft's framework builds and orchestrates agents, evolved from earlier Semantic Kernel and AutoGen work. It fits organisations already using Microsoft cloud services.

Google Agent Development Kit

Google's open-source kit for building and deploying agents, integrated with Gemini and Vertex AI.

Google's kit builds and deploys agents, integrated with Gemini and Vertex AI. It suits teams already on Google Cloud.

Production Agentic Systems

Reliable agents need evals, limits on steps and spend, observability, permissions, human review and rollback.

Reliable agents need evaluation, limits on steps and spend, observability, permissions, human review and rollback. A demo that works once is not a production agent.

Summary of terms

Term	Meaning
Introduction to Agentic AI	Systems where a model pursues a goal over several steps by planning, calling tools and reacting to results.
AI Agents vs Chatbots	A chatbot replies to messages; an agent takes actions toward a goal.
AI Agents vs Traditional Automation	Automation follows fixed rules; agents handle variable inputs and decide the next step, with less predictability.
Agent Architecture	A model, instructions, tools, memory, a control loop and guardrails.
Single-Agent Systems	One agent with a set of tools handles the task; simpler to build and debug.
Multi-Agent Systems	Several specialised agents coordinate, useful for separable roles but costlier and harder to control.
Agent Planning	Deciding the sequence of steps to reach a goal, revising as results arrive.
Reasoning and Task Decomposition	Breaking a goal into smaller sub-tasks the agent can complete and verify.
Tool Calling	The agent selects and invokes tools such as search, databases, code and APIs.
Function Calling	Structured model output naming a function and arguments, executed by your code.
Agent Memory	Information kept so the agent can use past steps, facts and preferences.
Short-Term and Long-Term Memory	Short-term lives in the current context; long-term is stored externally and retrieved when needed.
State Management	Tracking progress, inputs and results across steps so runs can resume and be audited.
Agent Orchestration	Coordinating agents, tools and steps, including routing, ordering and error paths.
Workflow Orchestration	Defining fixed or conditional flows around model calls for predictable behaviour.
Agent Communication	How agents exchange messages and shared state, including protocols such as MCP for tools and A2A for agents.
Human-in-the-Loop	Humans approve, edit or take over at defined checkpoints.
Autonomous Execution	Agents act without per-step approval, appropriate only for low-risk, reversible tasks.

Agent Permissions	Least-privilege access to tools and data, scoped per task and revocable.
Agent Error Handling	Detecting tool failures, bad outputs and loops, and responding safely.
Retry and Recovery Mechanisms	Retries with backoff, fallbacks, checkpoints and idempotent actions so failures do not cause harm.
Agent Evaluation	Measuring task success, steps, cost and safety on realistic scenarios.
Agent Observability	Traces of every model call, tool call and decision for debugging and audit.
Agent Security	Defending against prompt injection, data exfiltration, excessive permissions and malicious tools.
Agent Governance	Policies for what agents may do, approval rules, logging, ownership and incident response.
LangChain	A framework for building LLM applications with chains, tools and retrievers.
LangGraph	A library for stateful, graph-based agent workflows with loops and human checkpoints.
CrewAI	A framework for teams of role-based agents collaborating on tasks.
LlamaIndex	A framework centred on data ingestion, indexing and retrieval for LLM apps.
OpenAI Agents SDK	OpenAI's SDK for building agents with tools, handoffs and guardrails.
Microsoft Agent Framework	Microsoft's framework for building and orchestrating agents, evolved from Semantic Kernel and AutoGen.
Google Agent Development Kit	Google's open-source kit for building and deploying agents, integrated with Gemini and Vertex AI.
Production Agentic Systems	Reliable agents need evals, limits on steps and spend, observability, permissions, human review and rollback.

Practice exercise: Design an agent that books meetings. List its tools, permissions, approval points and what happens when a tool fails.

Self-check

1. Explain Introduction to Agentic AI in two sentences to a non-technical stakeholder, then say what you would measure for it.
2. Explain Human-in-the-Loop in two sentences to a non-technical stakeholder, then say what you would measure for it.
3. Explain Production Agentic Systems in two sentences to a non-technical stakeholder, then say what you would measure for it.

Related SkillNavora material

- How AI Agents Work: Single and multi-agent systems, context and evals.
- Agent Orchestration and Monitoring: Orchestrating and observing agents.
- AI Skills Academy: Every AI skills course in one place.
- AI Tutor: Ask questions on any topic.
- Study Library: Every lesson, class and PDF by programme.

© SkillNavora · skillnavora.in · Facts and tools change; check official sites before relying on a figure.